

Notes: Bloom, Sadun, and Van Reenen (AER, 2012)
Americans Do IT Better: US Multinationals and the Productivity Miracle

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1 Big picture

- **Question.** Why did the United States experience a post-1995 productivity acceleration while Europe did not, especially in sectors that intensively use information technology (IT)?
- **Core hypothesis.** The paper argues that US firms have organizational and managerial practices that make them better at extracting productivity gains from IT.
- **Key empirical idea.** If the advantage is not only the US environment but also US firm organization, then US multinationals operating in Europe should also obtain higher productivity from IT than non-US multinationals operating in the same European environment.
- **Main evidence, UK data.** In UK establishment-level panel data, US-owned establishments have a higher output elasticity of IT than non-US multinationals and domestic UK firms, especially in IT-using intensive sectors.
- **Selection test.** Establishments taken over by US multinationals do not have unusually high IT productivity before takeover, but they show higher IT productivity after takeover, especially after two or three years.
- **Mechanism evidence.** In a separate European firm-level panel, US firms have stronger people-management practices. These practices are complementary with IT and largely explain the US IT-productivity advantage.
- **People management.** The relevant management practices involve hiring, rewarding, promoting, and removing workers. These practices help firms reorganize work and exploit IT-enabled changes.
- **Quantitative implication.** The paper's back-of-the-envelope calculation suggests that the stronger US IT-productivity relationship can account for about half of the US-Europe productivity growth gap after 1995.
- **Economic takeaway.** The productivity miracle is not just about buying computers. It is about combining IT with complementary management practices, especially the tougher people-management style used by US firms.

2 Incremental contribution of the paper

- **From aggregate IT facts to micro mechanisms.** Earlier work documents that US productivity accelerated after 1995, especially in IT-using sectors. This paper uses micro data to ask why US firms used IT more productively.
- **A multinational test of the management hypothesis.** The paper separates the US environment from US firm organization by studying US multinationals operating in Europe. If US-owned affiliates in Europe also use IT better, the explanation cannot be only US geography, regulation, or local market conditions.
- **Two complementary micro panels.** The first data set is a UK establishment panel with detailed IT investment, output, inputs, and ownership. The second is a seven-country European firm panel combining IT data, accounting data, and a direct management survey.

- **IT as a complementary input.** The paper does not merely estimate an average return to IT. It allows the IT coefficient to vary by ownership type and by management practices.
- **Takeover design.** By examining establishments before and after acquisition by US versus non-US multinationals, the paper directly addresses the concern that US firms simply cherry-pick high-IT-productivity establishments.
- **Direct measurement of organization.** The paper uses survey-based measures of people management to show that the previously unobserved US advantage is linked to concrete practices: hiring, firing, promotion, and rewards.
- **Sectoral link to the productivity miracle.** The US advantage is concentrated in the IT-using sectors that macro studies identify as central to the post-1995 productivity acceleration.
- **Broader implication for intangible capital.** The paper gives empirical content to the idea that intangible organizational capital is crucial for realizing the productivity benefits of new technologies.

3 Data and measurement

3.1 UK establishment-level panel

- The UK panel is built by matching the Annual Business Inquiry with UK Census Bureau IT surveys and ownership information.
- The sample covers 1995-2003 and includes both manufacturing and services, excluding financial services.
- The data contain detailed establishment-level measures of gross output, employment, materials, non-IT capital, IT investment, and ownership.
- The authors construct IT capital stocks from IT expenditure flows using a perpetual inventory method with assumptions on depreciation and hedonic prices.
- The final UK panel contains 21,746 establishment-year observations with positive values for all factor inputs.
- Ownership is classified into three categories: US multinationals, non-US multinationals, and UK domestic firms.

Interpretation: the UK is a useful laboratory because it has many foreign-owned establishments, frequent ownership changes, and census-level IT and productivity data.

3.2 European firm-level panel

- The second panel combines three sources: the CEP management survey, Harte-Hanks IT data, and AMADEUS accounting data.
- The sample covers firms in France, Germany, Italy, Poland, Portugal, Sweden, and the United Kingdom.
- The panel used in the main management regressions includes 720 firms and covers 1999-2006.

- IT intensity is measured as computers per worker, aggregated from establishment-level Harte-Hanks data to the firm level.
- Accounting data provide sales, employment, and fixed assets.
- The panel also includes information on the share of workers with college degrees, allowing tests of whether the result is only skill-IT complementarity.

Interpretation: the European panel is smaller than the UK establishment panel, but it directly observes management practices, which allows the paper to test the mechanism.

3.3 People-management practices

The paper focuses on four people-management questions from the Bloom and Van Reenen management survey:

- managing human capital;
- rewarding high performance;
- removing poor performers;
- promoting high performers.

The four questions are normalized and averaged into a people-management z-score. Higher scores reflect a more performance-oriented approach to hiring, firing, promotion, and pay.

Interpretation: the paper does not assume these practices are universally better in every context. The claim is narrower: these US-style people-management practices are complementary with IT.

3.4 IT-using intensive sectors

The paper splits industries into IT-using intensive sectors and other sectors following Stiroh's classification. IT-using intensive sectors include many market service industries such as retail, wholesale, and business services, as well as some high-tech manufacturing sectors.

Interpretation: this split is central because the US productivity acceleration after 1995 occurred largely in IT-using sectors rather than only in IT-producing sectors.

4 Theory and empirical specifications

4.1 Production function with IT-management complementarity

The model starts from a production function in which IT capital and organizational capital are complementary:

$$Q_{it} = A_{it} O_{it}^{\alpha_O} C_{it}^{\alpha_C + \sigma O_{it}} L_{it}^{\alpha_L - \sigma O_{it}} K_{it}^{\alpha_K} M_{it}^{\alpha_M}.$$

Here Q is output, C is IT capital, O is organization or management, L is labor, K is non-IT capital, and M is materials. The key parameter is σ . If $\sigma > 0$, then better organization raises the productivity of IT.

Interpretation: the model formalizes the idea that computers by themselves are not enough. IT becomes powerful when paired with complementary organizational practices.

4.2 Estimating IT productivity when management is unobserved

When organization O is unobserved, the paper estimates production functions that allow the IT coefficient to differ by ownership status:

$$\begin{aligned}(q - l)_{it} = & \alpha_{C,DOM}(c - l)_{it} + \alpha_K(k - l)_{it} + \alpha_M(m - l)_{it} \\ & + (\alpha_C + \alpha_L + \alpha_M + \alpha_K - 1)l_{it} \\ & + \alpha_{C,USA}[(c - l)D^{USA}]_{it} + \alpha_{C,MNE}[(c - l)D^{MNE}]_{it} \\ & + a_i + \delta_{USA}D_{it}^{USA} + \delta_{MNE}D_{it}^{MNE} + \gamma'z_{it} + \xi_{kt} + u_{it}.\end{aligned}$$

The omitted ownership category is domestic firms. The central prediction is:

$$\alpha_{C,USA} > \alpha_{C,MNE} \geq \alpha_{C,DOM}.$$

Interpretation: if US multinationals have better complementary management, their output should respond more to IT capital.

4.3 IT intensity equation

The complementarity hypothesis also predicts that firms with the relevant organizational practices should choose higher IT intensity. The paper estimates:

$$(c - l)_{it} = \beta_{USA}D_{it}^{USA} + \beta_{MNE}D_{it}^{MNE} + \phi'w_{it} + \varsigma_{kt} + e_{it}.$$

The prediction is:

$$\beta_{USA} > \beta_{MNE} > 0.$$

Interpretation: if US firms are especially good at using IT, they should both obtain higher productivity from IT and use more IT.

4.4 Direct management test

In the European panel, management is observed directly. The paper estimates production functions that include people management and its interaction with IT:

$$(q - l)_{it} = \dots + \sigma[(c - l)_{it} \times O_i] + \alpha_O O_i + \alpha_{C,USA}[(c - l)D^{USA}]_{it} + \alpha_{C,MNE}[(c - l)D^{MNE}]_{it} + \dots.$$

If the US advantage comes from people-management practices, then the coefficient on $IT \times US$ should shrink and become insignificant once $IT \times PeopleManagement$ is included.

Interpretation: this is the key mechanism test. It asks whether measured people management absorbs the US ownership advantage in IT productivity.

4.5 Takeover specification

To address selection, the paper studies UK establishments that are acquired during the sample period. It compares establishments acquired by US multinationals with those acquired by non-US multinationals, before and after takeover.

The crucial test is whether $US \times IT$ is insignificant before takeover but positive after takeover.

Interpretation: this design checks whether US acquirers select establishments that already had high IT productivity or whether the US ownership effect emerges after management transfer and reorganization.

5 Identification and empirical design

5.1 What the design can identify

The paper identifies whether US-owned establishments in Europe have a higher association between IT and productivity than comparable establishments owned by non-US multinationals or domestic firms. Because all firms operate in European environments, the comparison helps isolate the role of firm organization from the role of the US domestic environment.

Interpretation: the design is strongest when comparing US multinationals to non-US multinationals, because both groups operate abroad and are likely to be more comparable than domestic firms.

5.2 Main identification challenges

- **Home-environment advantage.** US firms in the United States may look better because they operate at home. The paper addresses this by studying US multinationals in Europe.
- **Selection or cherry-picking.** US multinationals may acquire establishments with higher latent IT productivity. The takeover analysis checks pre-acquisition patterns.
- **Unobserved establishment heterogeneity.** Establishment fixed effects control for time-invariant omitted variables.
- **Input endogeneity.** The paper reports robustness using Olley-Pakes style methods and System GMM.
- **Transfer pricing and markups.** The paper checks whether US firms simply report different prices or accounting profits by interacting ownership with non-IT inputs and using markup-style controls.
- **IT measurement error.** The paper uses alternative IT measures, including the share of workers using computers.
- **Skill-IT complementarity.** The European panel controls for college-educated workers and interacts skills with IT. The people-management interaction remains important.

Interpretation: the paper does not rely on one single regression. Its credibility comes from the same pattern appearing across production functions, IT intensity equations, takeover evidence, and direct management measures.

5.3 Why people management is a plausible mechanism

IT often changes tasks, decision rights, monitoring, and the allocation of workers. Effective IT adoption may require shedding obsolete jobs, hiring different workers, retraining employees, giving high performers stronger incentives, and removing underperformers. These changes are precisely the kinds of organizational adjustments captured by the people-management score.

Interpretation: the mechanism is not that US firms simply have more computers. It is that US firms reorganize people around computers more effectively.

6 Tables and figures

6.1 Figures 1 and 2: the macro puzzle

Read:

- Figure 1 shows that output per hour in the United States and Europe rose together through much of the 1980s and early 1990s, but the United States pulled ahead after 1995.
- Figure 2 shows a parallel gap in IT capital per hour. US IT capital per hour rose sharply after 1995 and remained far above Europe.
- The two figures motivate the paper's puzzle: IT was available internationally, but the productivity gains from IT were much stronger in the United States.
- The paper focuses on why IT-using sectors in Europe did not experience the same acceleration as the United States.

Economic message: The macro evidence suggests that the key issue is not access to IT itself, but the ability to use IT productively.

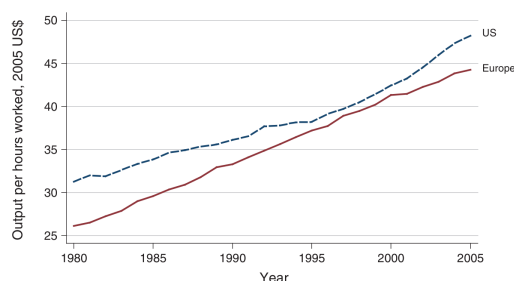


FIGURE 1. OUTPUT PER HOUR IN EUROPE AND THE UNITED STATES, 1980–2005

(a) Figure 1: output per hour in Europe and the United States

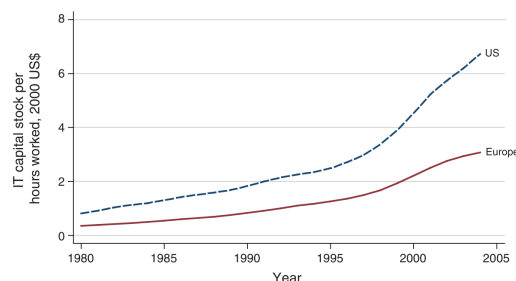


FIGURE 2. IT CAPITAL PER HOUR IN EUROPE AND THE UNITED STATES, 1980–2005

(b) Figure 2: IT capital per hour in Europe and the United States

6.2 Tables 1 and 2: UK descriptive facts and main production-function evidence

Read:

- Table 1 compares UK establishments by ownership status, normalized relative to the three-digit industry-year average.
- US multinationals are larger and more productive than other multinationals and domestic UK firms.
- The largest input-intensity gap is for IT capital per employee: US multinationals have a mean of 152.13, compared with 119.58 for other multinationals and 83.95 for UK domestic firms.
- Table 2 estimates production functions allowing the IT coefficient to differ by ownership.
- In the full sample, $US \times IT$ is positive and significant, whereas $MNE \times IT$ is small.
- The key result is strongest in IT-using intensive sectors. In column (7), with establishment fixed effects, $US \times \ln(C/L) = 0.0368$, significant at 1 percent, while $MNE \times \ln(C/L)$ is essentially zero.
- The p-value testing equality of US and non-US multinational IT coefficients in column (7) is 0.0094.

Economic message: US multinationals in the UK are more IT intensive and obtain higher productivity from IT, especially in sectors where IT should matter most.

TABLE 1—UK DESCRIPTIVE STATISTICS BROKEN DOWN BY MULTINATIONAL STATUS
(normalized to 100 for the 3-digit SIC and year average)

	Employment	Value added per employee	Gross output per employee	Non-IT capital per employee	Materials per employee	IT capital per employee
US multinationals						
Mean	162.26	127.96	123.63	129.61	123.81	152.13
Standard deviation	297.58	163.17	104.81	133.91	123.35	234.41
Observations	569	569	569	569	569	569
Other multinationals						
Mean	148.58	113.71	115.22	120.65	116.02	119.58
Standard deviation	246.35	107.87	86.50	126.83	107.63	180.34
Observations	2,119	2,119	2,119	2,119	2,119	2,119
UK domestic						
Mean	68.78	89.86	89.69	86.33	89.29	83.95
Standard deviation	137.72	104.50	102.09	127.16	129.37	188.30
Observations	4,433	4,433	4,433	4,433	4,433	4,433

Note: These are 2001 values from our sample of 7,121 establishments in the UK data (ABI matched with IT data from QICE, BSCI, and FAR).

(a) Table 1: UK descriptive statistics by multinational status

TABLE 2—ESTIMATES OF THE UK PRODUCTION FUNCTION ALLOWING THE IT COEFFICIENT TO DIFFER BY OWNERSHIP STATUS

Dependent variable:	ln(Q/L) (1)	ln(Q/L) (2)	ln(Q/L) (3)	ln(Q/L) (4)	ln(Q/L) (5)	ln(Q/L) (6)	ln(Q/L) (7)	ln(Q/L) (8)
Sectors	All sectors	All sectors	All sectors	IT using intensive sectors	Other sectors	All sectors	IT using intensive sectors	Other sectors
Fixed effects	NO	NO	NO	NO	NO	YES	YES	YES
USA $\times \ln(C/L)$			0.0202*** (0.0072)	0.0380*** (0.0128)	0.0120 (0.0084)	0.0093 (0.0085)	0.0368*** (0.0144)	-0.0060 (0.0098)
USA ownership $\times$ IT capital per employee								
MNE $\times \ln(C/L)$			0.0036 (0.0045)	-0.0011 (0.0062)	0.0062 (0.0060)	0.0010 (0.0042)	-0.0003 (0.0064)	0.0008 (0.0053)
Non-US multinational $\times$ IT capital per employee								
ln(C/L)		0.0457*** (0.0024)	0.0428*** (0.0029)	0.0373*** (0.0038)	0.0457*** (0.0039)	0.0152*** (0.0030)	0.0123** (0.0051)	0.0157*** (0.0036)
IT capital per employee								
ln(M/L)	0.5575*** (0.0084)	0.5474*** (0.0083)	0.5477*** (0.0083)	0.6216*** (0.0142)	0.5067*** (0.0104)	0.4031*** (0.0178)	0.5018*** (0.0279)	0.3606*** (0.0210)
Materials per employee								
ln(K/L)	0.1388*** (0.0071)	0.1268*** (0.0068)	0.1268*** (0.0068)	0.1106*** (0.0093)	0.1459*** (0.0092)	0.0900*** (0.0159)	0.1056*** (0.0228)	0.0666*** (0.0209)
Non-IT capital per employee								
ln(L)	-0.0052* (0.0027)	-0.0112*** (0.0027)	-0.0111*** (0.0027)	-0.0094** (0.0037)	-0.0121*** (0.0036)	-0.1986*** (0.0217)	-0.1279*** (0.0319)	-0.2466*** (0.0279)
Labor								
USA	0.0711*** (0.0140)	0.0641*** (0.0135)	0.0733*** (0.0144)	0.0440* (0.0213)	0.0892*** (0.0189)	0.0214 (0.0224)	0.0451 (0.0366)	-0.0070 (0.0242)
USA ownership								
MNE	0.0392*** (0.0079)	0.0339*** (0.0078)	0.0372*** (0.0093)	0.0149 (0.0134)	0.0441*** (0.0124)	0.0081 (0.0103)	0.0173 (0.0172)	-0.0008 (0.0126)
Non-US multinational								
Observations	21,746	21,746	21,746	7,784	13,962	21,746	7,784	13,962
Test USA $\times \ln(C/L) = MNE \times \ln(C/L)$			0.0320	0.0035	0.5272	0.3622	0.0094	0.5210
Test USA = MNE, p-value		0.0206	0.0232	0.0113	0.1755	0.5545	0.4301	0.8145

Notes: The dependent variable in all columns is the log of gross output per employee. The time period is 1995–2003. The estimation method in all columns is OLS. Columns 6 to 8 include establishment-level fixed effects. Standard errors in brackets under coefficients in all columns are clustered by establishment (i.e., robust to heteroskedasticity and autocorrelation of unknown form). All columns include a full set of three-digit industry dummies interacted with a full set of time dummies and as additional controls: dummies for establishment age (interacted with a manufacturing dummy), region, multiestablishment group (interacted with ownership type), and a dummy for IT survey. See online Table A1 for definition of IT using intensive sectors. "Test USA $\times \ln(C/L) = MNE \times \ln(C/L)$ " is a test of whether the coefficient on USA $\times \ln(C/L)$ is significantly different from the coefficient on MNE $\times \ln(C/L)$, etc.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

(b) Table 2: UK production function and ownership-specific IT coefficients

6.3 Tables 3 and 4: robustness and IT demand

Read:

- Table 3 focuses on IT-intensive sectors and tests whether the main production-function result is robust.

- The $US \times IT$ interaction remains positive and significant when the dependent variable is value added, when all inputs are interacted with ownership, when an alternative IT measure is used, when translog interactions are added, and when non-US multinationals are split into EU and non-EU groups.
- The result is not explained by transfer pricing or general markup differences because the US ownership interactions with non-IT inputs are not jointly significant.
- Table 4 estimates IT intensity equations.
- US-owned establishments have significantly higher IT capital per worker than non-US multinationals and domestic firms.
- In all sectors, the US coefficient is 0.2629 without additional controls and 0.2406 with additional controls. The tests of equality between US and non-US multinationals are significant or marginally significant.

Economic message: The same ownership pattern appears in two margins: US firms both use more IT and obtain more productivity from IT.

TABLE 3—ROBUSTNESS TESTS OF THE UK PRODUCTION FUNCTION

	(1)	(2)	(3)	(4)	(5)	(6)
Experiment	Baseline specification	Value added	All inputs interacted	Alternative IT measure	Full "translog" interactions	EU and non-EU MNEs
Dependent variable:	$\ln(Q/L)$	$\ln(VA/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$
USA $\times \ln(C/L)$	0.0368**	0.0681**	0.0328**	0.0672**	0.0334**	0.0376***
USA ownership $\times$ IT capital per employee	(0.0144)	(0.0319)	(0.0141)	(0.0258)	(0.0140)	(0.0145)
MNE $\times \ln(C/L)$	-0.0003	-0.0179	0.0002	0.0070	-0.0012	
Non-US multinational $\times$ IT capital per employee	(0.0064)	(0.0166)	(0.0065)	(0.0126)	(0.0062)	
$\ln(C/L)$	0.0123**	0.0290***	0.0126**	0.0262***	0.0330	0.0120**
IT capital per employee	(0.0051)	(0.0110)	(0.0050)	(0.0082)	(0.0460)	(0.0051)
USA $\times \ln(M/L)$			0.0334			
USA ownership $\times$ materials per employee			(0.0376)			
MNE $\times \ln(M/L)$			0.0080			
Non-US multinational $\times$ materials per employee			(0.0236)			
USA $\times \ln(K/L)$			0.0241			
USA ownership $\times$ non-IT capital per employee			(0.0368)			
MNE $\times \ln(K/L)$			-0.0142			
Non-US $\times$ non-IT capital per employee			(0.0134)			
EU MNE						0.0063
EU ownership						(0.0198)
Non-EU MNE						-0.0603
Non-EU-non-USA ownership						(0.0489)
EU MNE $\times \ln(C/L)$						0.0016
EU ownership $\times$ IT capital per employee						(0.0064)
Non-EU MNE $\times \ln(C/L)$						-0.0140
Non-EU-non-USA $\times$ IT capital per employee						(0.0157)
Observations	7,784	7,784	7,784	2,155	7,784	7,784
Test USA $\times \ln(C) =$ MNE $\times \ln(C)$, p -value	0.0094	0.0103	0.0224	0.0216	0.0138	
Test USA = MNE, p -value	0.4301	0.9638	0.3620	0.2244	0.3852	
Test on joint significance of all the interaction terms, excluding IT interactions (p -value)			0.3752			
Test on joint significance of all the US interaction terms, excluding IT per employee (p -value)			0.6216			
Test on all the other MNE's interaction terms, excluding IT per employee (p -value)			0.2723			
Test on additional "translog" terms, p -value					0.0000	
Test USA = EU, p -value						0.3216
Test USA = non-EU, p -value						0.0815
Test [USA $\times \ln(C/L)$] = [EU $\times \ln(C/L)$], p -value						0.0120
Test [USA $\times \ln(C/L)$] = [non-EU $\times \ln(C/L)$], p -value						0.0123

Notes: The dependent variable in all columns is the log of gross output per employee, except (2) which is value added per employee. All columns are for only the sectors that use IT intensively (see online Appendix A1). The time period is 1995–2003. The estimation method is OLS. All columns also include (the log of) non-IT capital per worker (K/L), materials per worker (M/L) and labor (L). All columns except 4 include establishment fixed effects. Standard errors in brackets under coefficients are clustered by establishment (i.e., robust to heteroskedasticity and autocorrelation of unknown form). All columns include a full set of three-digit industry dummies interacted with a full set of time dummies and as additional controls: dummies for establishment age (interacted with a manufacturing sector dummy), region, multiestablishment group (interacted with ownership type), and IT survey (except column 4). The IT measure in column 4 is the $\ln$ fraction of workers using computers. Column 5 includes all the pairwise interactions of materials, labor, IT capital, and non-IT capital, and the square of each of these factors. *Test USA $\times \ln(C/L) =$ MNE $\times \ln(C/L)$ is a test of whether the coefficient on USA $\times \ln(C/L)$ is significantly different from the coefficient on MNE $\times \ln(C/L)$, etc.

***Significant at the 1 percent level.

(a) Table 3: robustness tests of the UK production function

TABLE 4—UK IT INTENSITY EQUATIONS

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	$\ln(C/L)$	$\ln(C/L)$	$\ln(C/L)$	$\ln(C/L)$	$\ln(C/L)$	$\ln(C/L)$
Sectors	All intensive sectors	IT-using intensive sectors	Other sectors	All intensive sectors	IT-using intensive sectors	Other sectors
USA	0.2629***	0.3393***	0.2085***	0.2406***	0.3129***	0.1927***
USA ownership	(0.0461)	(0.0717)	(0.0600)	(0.0463)	(0.0717)	(0.0604)
MNE	0.1632***	0.2117***	0.1332***	0.1506***	0.1939***	0.1228***
Non-US multinational	(0.0287)	(0.0440)	(0.0375)	(0.0291)	(0.0452)	(0.0380)
Additional controls	NO	NO	NO	YES	YES	YES
Observations	21,746	7,784	13,962	21,746	7,784	13,962
Test USA = MNE, p -value	0.0310	0.0758	0.2108	0.0528	0.0970	0.2508

Notes: The dependent variable in all columns is the log of IT capital per employee. The time period is 1995–2003. The estimation method in all columns is OLS. Standard errors in brackets under coefficients in all columns are clustered by establishment (i.e., robust to heteroskedasticity and autocorrelation of unknown form). All columns include a full set of three-digit industry dummies interacted with a full set of time dummies and the log of gross output. Additional controls include dummies for establishment age (interacted with a manufacturing dummy), region, multi-establishment group (interacted with ownership type) and IT survey. See online Table A1 for definition of IT using intensive sectors. *Test USA = MNE is test of whether the coefficient on USA is significantly different from the coefficient on MNE.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

(b) Table 4: UK IT intensity equations

6.4 Table 5: production functions before and after takeovers

Read:

- Table 5 restricts attention to establishments that are taken over at some point in the sample.
- Before takeover, establishments that will be acquired by US multinationals do not have significantly higher IT productivity. The $US \times IT$ coefficient is negative and insignificant.
- After takeover, the $US \times IT$ coefficient becomes positive and significant. In column (4), it is 0.0541 and significant at 5 percent.
- The post-takeover US effect appears with a lag. In column (5), the $US \times IT$ coefficient is insignificant one year after takeover but positive and significant two and three years after takeover.
- Dropping UK domestic acquirers in column (6) yields a similar pattern: the two-to-three-year post-takeover interaction is 0.1303 and significant at 5 percent.

Economic message: The takeover evidence is hard to reconcile with simple cherry-picking. The US IT-productivity advantage emerges after acquisition, consistent with management transfer and reorganization.

TABLE 5—UK PRODUCTION FUNCTIONS BEFORE AND AFTER TAKEOVERS

	(1)	(2)	(3)	(4)	(5)	(6)
Sample	Before takeover	Before takeover	After takeover	After takeover	After takeover	After takeover (drop UK domestic acquirers)
Dependent variable: $\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$
USA $\times \ln(C/L)$		−0.0672 (0.0749)		0.0541** (0.0273)		
USA takeover $\times$ IT capital per employee						
MNE $\times \ln(C/L)$		−0.0432 (0.0463)		0.0073 (0.0150)		
Non-US multinational takeover $\times$ IT capital per employee						
USA	−0.0661 (0.0663)	−0.1055 (0.0863)	0.0353 (0.0402)	0.0619 (0.0461)		
USA takeover						
MNE	0.0321 (0.0565)	−0.0009 (0.0710)	0.0117 (0.0298)	0.0205 (0.0342)		
Non-US multinational takeover						
USA $\times \ln(C/L)$ one year after takeover					0.0192 (0.0378)	0.0191 (0.0562)
USA $\times \ln(C/L)$ two and three years after takeover					0.0661** (0.0294)	0.1303** (0.0573)
MNE $\times \ln(C/L)$ one year after takeover					−0.0091 (0.0197)	
MNE $\times \ln(C/L)$ two and three years after takeover					0.0115 (0.0162)	
USA one year after takeover					0.0019 (0.0542)	0.0014 (0.0716)
USA two and three years after takeover					0.0934* (0.0485)	0.0942 (0.0856)
MNE one year after takeover					−0.0178 (0.0411)	
MNE two and three years after takeover					0.0327 (0.0361)	

(Continued)

TABLE 5—UK PRODUCTION FUNCTIONS BEFORE AND AFTER TAKEOVERS (Continued)

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(C/L)$	0.0744** (0.0299)	0.0935** (0.0432)	0.0395*** (0.0079)	0.0287*** (0.0088)	0.0288*** (0.0088)	0.0282 (0.0224)
IT capital per employee						
$\ln(M/L)$	0.5486*** (0.0489)	0.5487*** (0.0481)	0.6871*** (0.0173)	0.6892*** (0.0173)	0.6886*** (0.0172)	0.7323*** (0.0292)
Materials per employee						
$\ln(K/L)$	0.1759*** (0.0343)	0.1718*** (0.0335)	0.0350** (0.0160)	0.0350** (0.0159)	0.0353** (0.0159)	−0.0108 (0.0431)
Non-IT capital per employee						
$\ln(L)$	−0.0185 (0.0292)	−0.0215 (0.0276)	−0.0117 (0.0108)	−0.0111 (0.0108)	−0.0112 (0.0107)	−0.0358* (0.0213)
Labor						
Observations	261	261	1,006	1,006	1,006	241
Test USA $\times \ln(C/L)$ = MNE $\times \ln(C/L)$, p -value		0.7037		0.0965		
Test USA = MNE, p -value	0.1637	0.1773	0.5979	0.4056		
Test (USA one year) $\times \ln(C/L)$ = (MNE one year) $\times \ln(C/L)$, p -value					0.4948	
Test (USA two plus years) $\times \ln(C/L)$ = (MNE two plus years) $\times \ln(C/L)$, p -value						0.0734
Test USA one year = MNE one year, p -value						0.7463
Test USA two plus years = MNE two plus years, p -value						0.2481

Notes: The sample is all establishments in the IT-intensive sectors (online Table A1) who were taken over at some point (omitted base is “domestic takeovers”—UK firms taking over another firm). We drop takeovers that do not result in a change of ownership category (e.g., US takeovers of US firms, non-US MNE takeovers of non-US MNEs and domestic takeovers of domestic firms). The dependent variable is the log of gross output per employee. The time period is 1995–2003. The estimation method is OLS. Standard errors in brackets under coefficients are clustered by establishment. A takeover is defined as a change in the establishment foreign ownership marker or—for UK domestic establishment—as a change in the enterprise group marker. The “before” period is defined as the interval between one and three years before the takeover takes place. The “after” period is defined as the interval between one and three years after the takeover takes place. The year in which the takeover takes place is excluded from the sample. All columns include a full set of two-digit industry dummies interacted with time dummies and as additional controls: age, region dummies, a multiestablishment group dummy, an IT survey dummy, and controls for total takeover activity over the sample period. “Test USA $\times \ln(C/L)$ = MNE $\times \ln(C/L)$ ” is a test of whether the coefficient on USA $\times \ln(C/L)$ is significantly different from the coefficient on MNE $\times \ln(C/L)$, etc.

***Significant at the 1 percent level.
**Significant at the 5 percent level.

(a) Table 5, part 1: pre- and post-takeover ownership interactions

(b) Table 5, part 2: factor inputs, tests, and notes

6.5 Figure 3 and Table 6: people management as the mechanism

Read:

- Figure 3 shows that US firms have much higher people-management scores than firms in other countries.
- Panel A ranks firms by country of location; the United States is far above the rest.
- Panel B ranks European multinational subsidiaries by country of origin; US-owned subsidiaries have the highest people-management scores.
- Table 6 uses the European firm-level panel with direct management measures.

- Column (2) replicates the ownership-IT result: $US \times IT$ is 0.1790 and significant at 5 percent, while $MNE \times IT$ is negative and insignificant.
- Column (3) replaces ownership interactions with people management. $PeopleManagement \times IT$ is 0.1451 and significant at 1 percent.
- Column (4) includes both ownership-IT and management-IT interactions. The US ownership interaction falls by more than half and becomes insignificant, while $PeopleManagement \times IT$ remains positive and significant.
- Columns (7) and (8) show that US firms and firms with stronger people management also use more IT.

Economic message: The direct management data explain the source of the US advantage. US firms' stronger people-management practices are complementary with IT and account for most of the higher US IT-productivity coefficient.

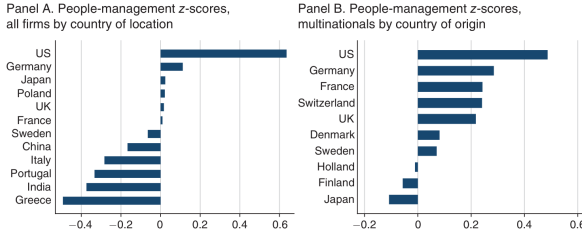


FIGURE 3

Notes: In Figure 3, panels A and B, the "People-management z-score" is the average z-score score for the 4 management practices on people management, covering "Managing human capital," "Rewarding high performance," "Removing poor performers," and "Promoting high performers." This is normalized to have a firm-level mean of zero standard deviation of 1. The sample in panel A is all 4,050 firms sorted according to country of location. The sample in panel B is the subset of 618 multinational subsidiaries located in France, Germany, Italy, Poland, Portugal, Sweden, and the United Kingdom, sorted according to country of origin and plotted only for origin coun-

TABLE 6—EUROPEAN FIRM-LEVEL PANEL DATA WITH DIRECT MEASURES OF MANAGEMENT

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(Q/L)$	$\ln(C/L)$	$\ln(C/L)$
Fixed effects	NO	NO	NO	NO	YES	YES	NO	NO
USA $\times \ln(C/L)$		0.1790** (0.0733)		0.0784 (0.0720)	0.0518 (0.0713)	0.0192 (0.0785)		
USA ownership $\times$ computers per employee								
MNE $\times \ln(C/L)$		-0.0263 (0.0586)		-0.0235 (0.0553)	0.0218 (0.0547)	0.0235 (0.0550)		
Non-US multinational $\times$ computers per employee								
People management			0.0271 (0.0219)	0.0271 (0.0219)				0.1268*** (0.0353)
People management $\times \ln(C/L)$			0.1451*** (0.0331)	0.1404*** (0.0344)	0.1284* (0.0773)	0.0994* (0.0581)		
People management $\times$ computers per employee								
$\ln(K/L)$	0.2401*** (0.0163)	0.1838*** (0.0284)	0.1782*** (0.0276)	0.1791*** (0.0276)	0.2347*** (0.0926)	0.2316*** (0.0882)		
Non IT capital per employee								
$\ln(L)$	-0.0182 (0.0162)	0.0421 (0.0360)	0.0421 (0.0344)	0.0409 (0.0349)	-0.2182 (0.2600)	-0.2347 (0.2497)		
Labor								
$\ln(C/L)$		0.1256*** (0.031)	0.1430*** (0.0284)	0.1463*** (0.0303)	-0.0493 (0.0596)	-0.2282 (0.1738)		
Computers per employee								
USA	0.2548*** (0.0438)	0.0779 (0.0481)	0.1111** (0.0446)	0.0837* (0.046)			0.2601*** (0.0742)	0.2150*** (0.0732)
USA ownership								
MNE	0.1909*** (0.0304)	0.1597*** (0.0363)	0.1604*** (0.0355)	0.1618*** (0.0357)			0.0492 (0.0596)	0.0367 (0.0591)
Non-US multinational								
$\ln(\text{degree})$		0.0433** (0.0183)	0.0375** (0.0184)	0.0370** (0.0184)			0.0585** (0.0293)	0.0359 (0.0296)
Percentage employees with a college degree								
$\ln(\text{degree}) \times \ln(C/L)$						0.0700 (0.0484)		
Percentage employees with a college degree $\times$ computers per employee								
Observations	9,463	2,555	2,555	2,555	2,555	2,555	2,555	2,555
Test USA $\times \ln(C/L) =$								
MNE $\times \ln(C/L)$, p-value		0.0189		0.2419	0.6360	0.9565		
Test USA = MNE, p-value	0.1789	0.1206	0.3094	0.1264			0.0095	0.0253

Notes: The dependent variable in columns 1 to 6 is the log of sales per employee, and in columns 7 and 8 is the log of computers per employee. The time period is 1999–2006, containing data from France, Germany, Italy, Poland, Portugal, Sweden, and the UK. The estimation method in all columns is OLS. Columns 5 and 6 include firm-level fixed effects. Standard errors in brackets under coefficients in all columns are clustered by firm (i.e., robust to heteroskedasticity and autocorrelation of unknown form). All columns include a full set of three-digit industry dummies, country dummies interacted with a full set of time dummies, and a public listing indicator. Columns 2 to 8 are weighted by the survey coverage rate in the Harte-Hanks data, plus include a fifth-order Taylor expansion for the coverage ratio to control for any potential survey bias. "Test USA $\times \ln(C/L) =$ MNE $\times \ln(C/L)$ " is a test of whether the coefficient on USA $\times \ln(C/L)$ is significantly different from the coefficient on MNE $\times \ln(C/L)$, etc. 720 firms in all columns except column 1 where there are 1,828 firms.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

(a) Figure 3: people-management scores

(b) Table 6: European firm-level panel with direct management measures

7 Short summary

- The paper studies the source of the US post-1995 IT-led productivity miracle.
- The key hypothesis is that US firms have management practices that make IT more productive.
- UK establishment data show that US multinationals operating in Europe obtain higher output from IT than non-US multinationals and domestic firms.

- This advantage is concentrated in IT-using intensive sectors, matching the sectors behind the aggregate productivity acceleration.
- The result survives establishment fixed effects, alternative IT measures, value-added specifications, translog controls, and input-endogeneity checks.
- Takeover evidence suggests that US firms do not merely select establishments with high preexisting IT productivity; the advantage appears after takeover.
- European firm-level data show that US firms have stronger people-management practices.
- People management is complementary with IT and largely absorbs the US ownership interaction with IT.
- US firms also use more IT, consistent with a higher marginal product of IT.
- The broader message is that IT productivity depends on complementary organizational capital.

8 Conclusion

8.1 Core conclusion

Bloom, Sadun, and Van Reenen show that US multinationals operating in Europe also experienced an IT-related productivity advantage. This suggests that the post-1995 US productivity miracle cannot be explained only by the US environment. A major part of the explanation lies in firm-level practices that US multinationals carry abroad.

8.2 Economic intuition

IT changes how work is organized. It can make information flows faster, allow better monitoring, and enable new processes, but these benefits require complementary organizational changes. Firms need to hire workers who can use new technologies, reward high performers, remove low performers, and reorganize tasks. US firms appear especially strong in these people-management practices, which makes IT more productive.

8.3 Incremental contribution revisited

The paper contributes by connecting a macro productivity puzzle to micro evidence on firms. It shows that the return to IT differs systematically by ownership and management practices. It also provides a credible mechanism by combining establishment-level production functions, takeover evidence, and direct survey measures of people management.

8.4 Implications

The findings imply that technology adoption is not just a capital investment decision. Policies or firm strategies that encourage IT investment may have limited effects unless firms also develop complementary management and organizational capabilities. The paper also suggests that multinational firms can transmit organizational know-how across borders.

8.5 Limitations

The evidence is observational, and the paper cannot fully randomize ownership, management practices, or IT investment. The management survey measures people-management practices only imperfectly. The strongest direct mechanism evidence comes from a smaller European firm panel rather than the larger UK establishment panel. In addition, the paper focuses on nonfinancial firms and largely on Europe, so the results may not transfer mechanically to every sector or region.

8.6 Takeaway

The enduring takeaway is that Americans do IT better because US firms combine IT with a distinctive form of organizational capital. Computers raised productivity most where firms had the people-management practices needed to reorganize work around them. The productivity miracle was therefore not only an IT story; it was an IT-plus-management story.